Exhibit B Drawings
RFP 27-02

GRANDVIEW ELEMENTARY SCHOOL
ROOF REPLACEMENT DRAWINGS

1583 GRAND AVE.
WINDSOR, CO

GENERAL ROOFING & CODE COMPLIANCE NOTES
GENERAL JURISDICTION & APPLICABLE CODES

Authority Having Jurisdiction (AHJ): Colorado Division of Fire Prevention and Control (DFPC). Applicable Codes: All work shall comply with the currently adopted editions of the DFPC regulatory framework, including but not limited to:
2021 International Building Code (IBC)
2021 International Existing Building Code (IEBC)
2021 International Energy Conservation Code (IECC)
2021 International Fire Code (IFC)
2023 Colorado Model Electric Ready and Solar Ready Code
ASCE/SEI 7-16 (American Society of Civil Engineers Standard Minimum Design Loads for Buildings and Other Structures) Occupancy & Construction: Building is classified as Group E (Educational) Occupancy, Risk Category III, Construction Type IIB.

ROOF DESIGN & PERFORMANCE CRITERIA
Fire Classification: The complete roof assembly (including EPDM and pedestal paver system) must achieve a Class A fire rating tested in accordance with UL 790 or ASTM E108. The Contractor shall provide the specific UL assembly number in the product submittals.
Thermal Resistance (IECC Zone 5B): The new roof assembly entirely above the deck shall provide a minimum continuous insulation value of R-30. Tapered polyiso layouts must be engineered to achieve this average.
Wind Uplift & Structural Criteria:
Location & Risk Category: Weld RE4 Grandview Elementary School (Windsor, CO), Risk Category III.
Basic Wind Speed (VULT): 115 mph (3-second gust), Exposure Category C per SEAC/DFPC criteria.
Historical Baseline: Legacy 1997 Uniform Building Code (UBC) calculations indicated a uniform uplift design pressure of 22 psf during all construction phases.
Current Code Design Pressure: Per 2021 IBC / ASCE 7-16, the required average ultimate wind uplift pressure for the new roof assembly is 16 psf (equivalent to 9.6 psf nominal, calculated as $16 psf X 0.6). Edge Flashing: All perimeter edge metal must be tested, certified, and installed in compliance with ANSI/SPRI ES-1 standards.

PERMITTING & ADMINISTRATIVE SUBMITTALS
DFPC Documentation: The Contractor is responsible for submitting all required project documentation directly to Safebuilt, the pre-qualified school district building department.



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GENERAL NOTES:

All roofing products are to be installed according to manufacturer's recommendations and the project performance requirements in the specification.

Dimensions are to be field verified (V.I.F.) and dimensions on roof plan (provided from architectural drawings) and details are not to scale (N.T.S.).

Review all conditions prior to installation and make revisions as needed to ensure proper drainage and for compliance with roof system manufacturer's warranty requirements.

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ROOF SYMBOL LEGEND		DETAIL
	CURB	4
	PIPE	5
RD	ROOF DRAIN	6
	ROOF HATCH	7
S/DS	SCUPPER	8
OS	OVERFLOW SCUPPER	9
	WALKPAD	N/A

ALTERNATE BID
ADHERED SYSTEM -
WALKPADS ONLY AROUND UNITS, ROOF HATCHES,
AND LADDERS

BASE BID
BALLASTED SYSTEM -
WALKWAY PER LAYOUT/EPDM SLIP SHEET
& CONCRETE PAVER

ROOF PLAN

SCALE 1/16" = 1'-0"

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GENERAL NOTES:

-ALL INSULATION LAYER JOINTS TO BE STAGGERED AND OFFSET A MINIMUM OF 6" IN ALL DIRECTIONS FROM EACH SUCCEEDING LAYER.

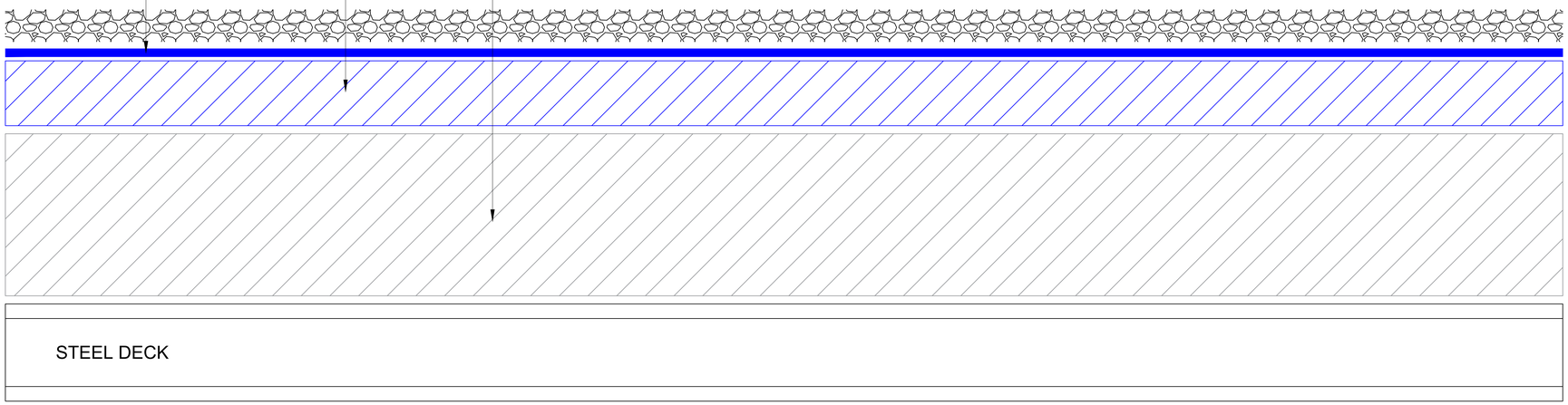
-THE SURFACE TO WHICH INSULATION IS TO BE APPLIED SHALL BE DRY, FREE OF FINS, PROTRUSIONS, SHARP EDGES, LOOSE AND FOREIGN MATERIALS, OIL AND GREASE. AREA SHOULD BE CLEANED WITH AIR BLOWER.

-REVIEW ALL CONDITIONS PRIOR TO INSTALLATION AND MAKE REVISIONS AS NEEDED TO ENSURE PROPER DRAINAGE AND FOR COMPLIANCE WITH ROOF SYSTEM MANUFACTURER'S WARRANTY REQUIREMENTS.

90-MIL EPDM MEMBRANE;
LOOSE LAID AND HELD IN PLACE WITH 1-1/2" NOMINAL ROUND WATER WORN GRAVEL CONFORMING TO GRADATION #4 IN ACCORDANCE WITH ASTM D-448 SIZING. WITH COVERAGE MINIMUM OF:
- 10 LBS.PER SQUARE FOOT (10' PERIMETER)
- 13 LBS PER SQUARE FOOT OF #2 (2-1/2") STONE (10' X 10' CORNER)

MIN. 1.0" 20-PSI INSULBASE POLYISO INSULATION - LOOSE LAID

EXISTING INSULATION TO REMAIN;
-REMOVE AND REPLACE ANY COMPROMISED, DAMAGED, OR WET AREAS OF EXISTING



ASSEMBLY SECTION CUT - BASE BID ASSEMBLY

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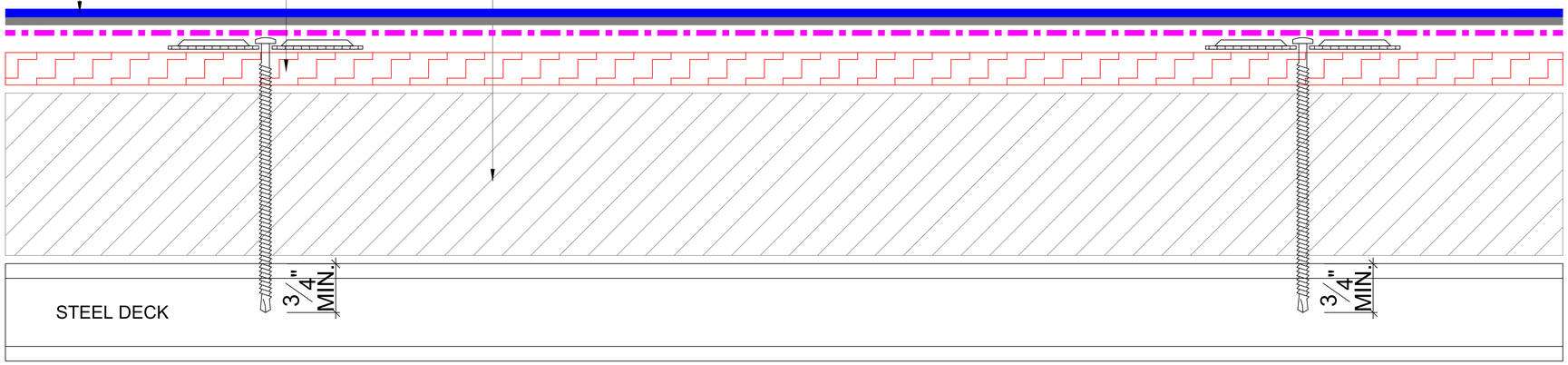
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145-MIL FLEECEBACK EPDM MEMBRANE; ADHERED WITH FLEXIBLE FAST AT THE FOLLOWING RATE:
- ALL ROOF ZONES: 4" O.C. BEADS

1/2" SECURSHIELD HD PLUS COVER BOARD (4' X 8' BOARD); MECHANICALLY FASTENED AT THE FOLLOWING RATE:
- ALL ROOF ZONES: 16 INSULFAST FASTENERS AND 3" INSULATION PLATES

EXISTING INSULATION TO REMAIN;
-REMOVE AND REPLACE ANY COMPROMISED, DAMAGED, OR WET AREAS OF EXISTING



ASSEMBLY SECTION CUT - ALTERNATE ASSEMBLY

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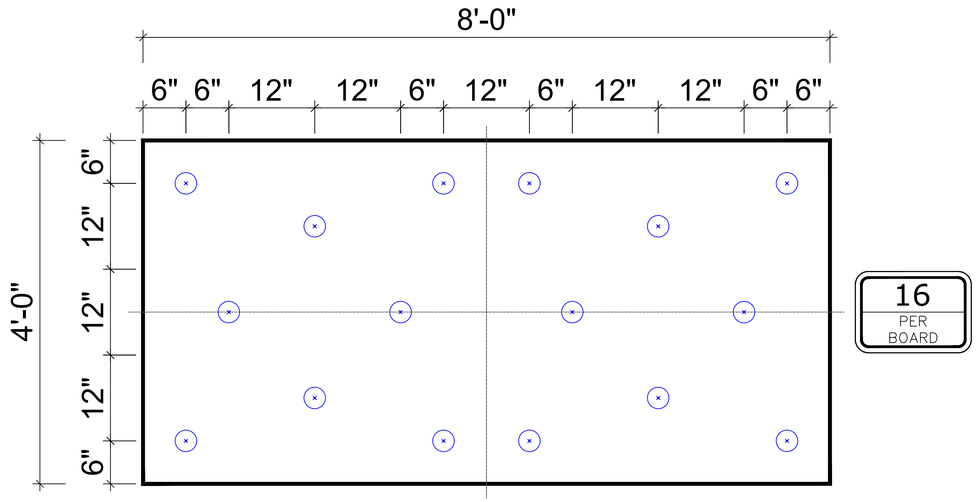
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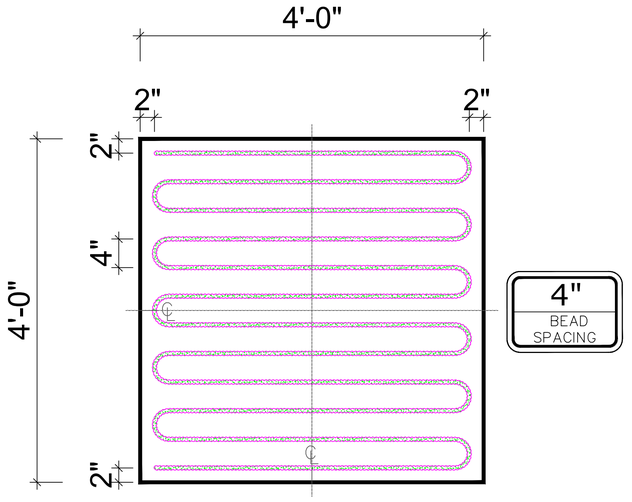
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1 INSULATION ATTACHMENT

16 PER 4'X8'



2 MEMBRANE ATTACHMENT

4" O.C. BEADS

COMPONENT ATTACHMENT

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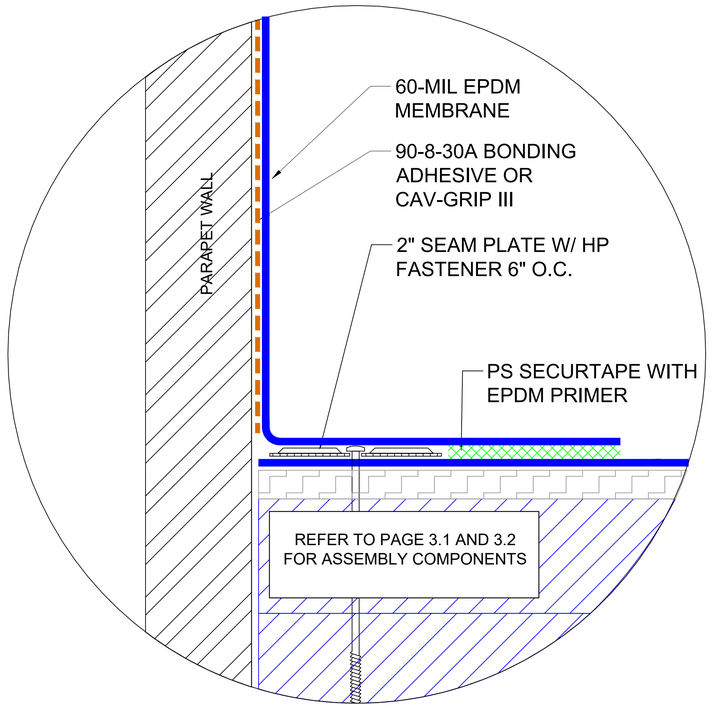
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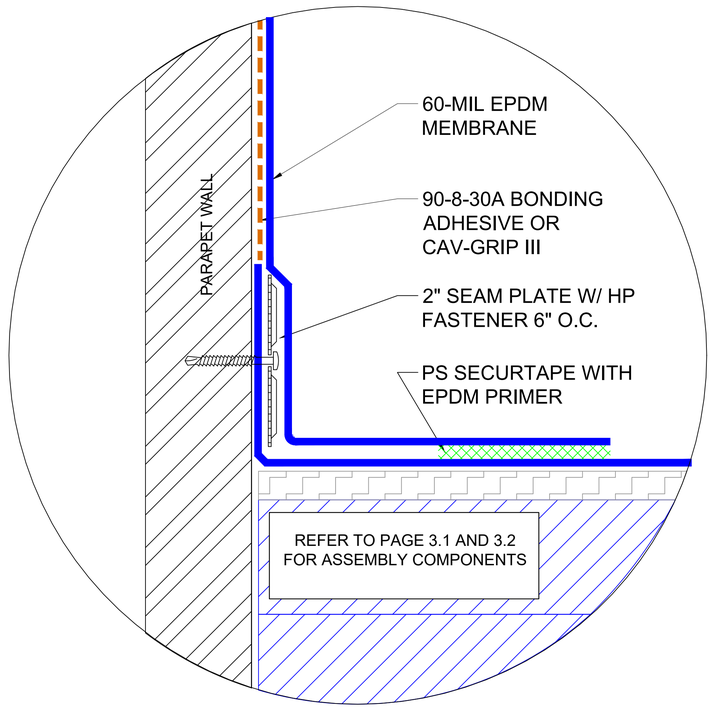
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SECUREMENT INTO DECK



SECUREMENT INTO VERTICAL SUBSTRATE

MEMBRANE ATTACHMENT

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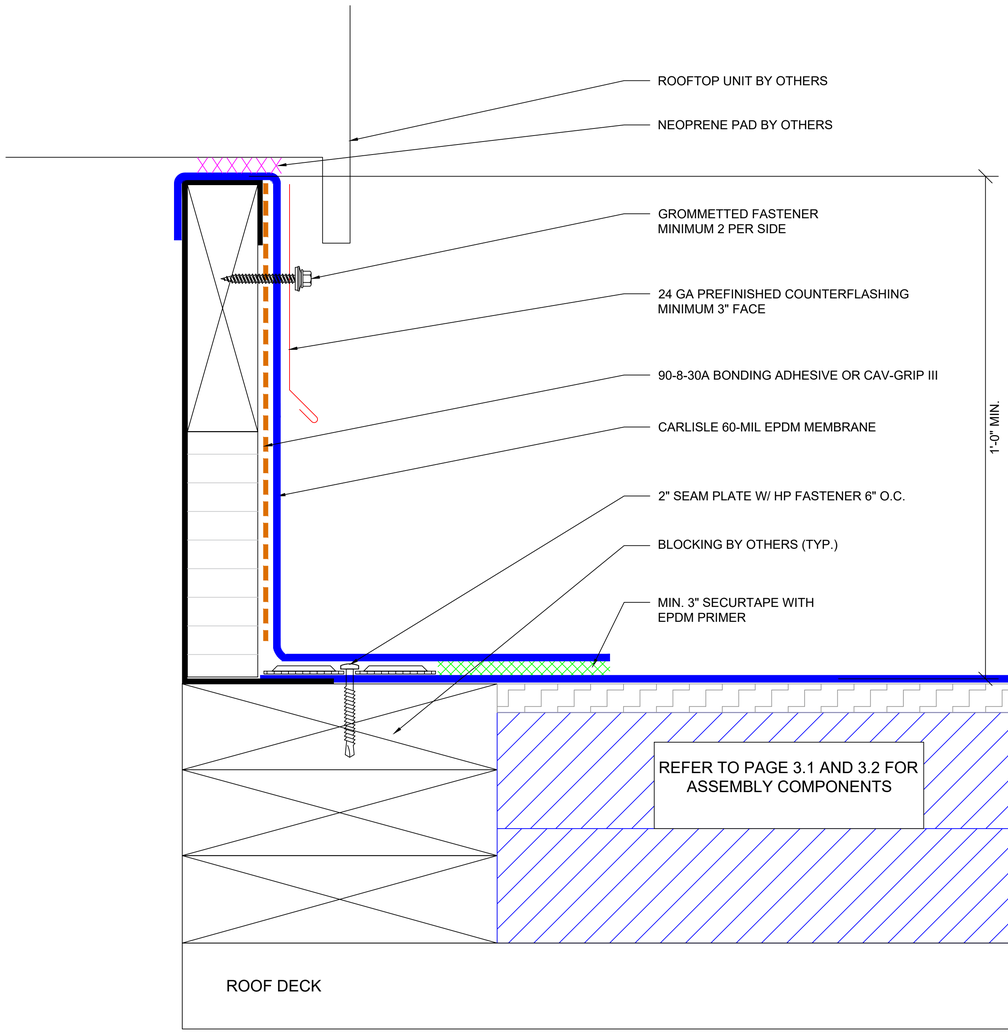
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EQUIPMENT CURB

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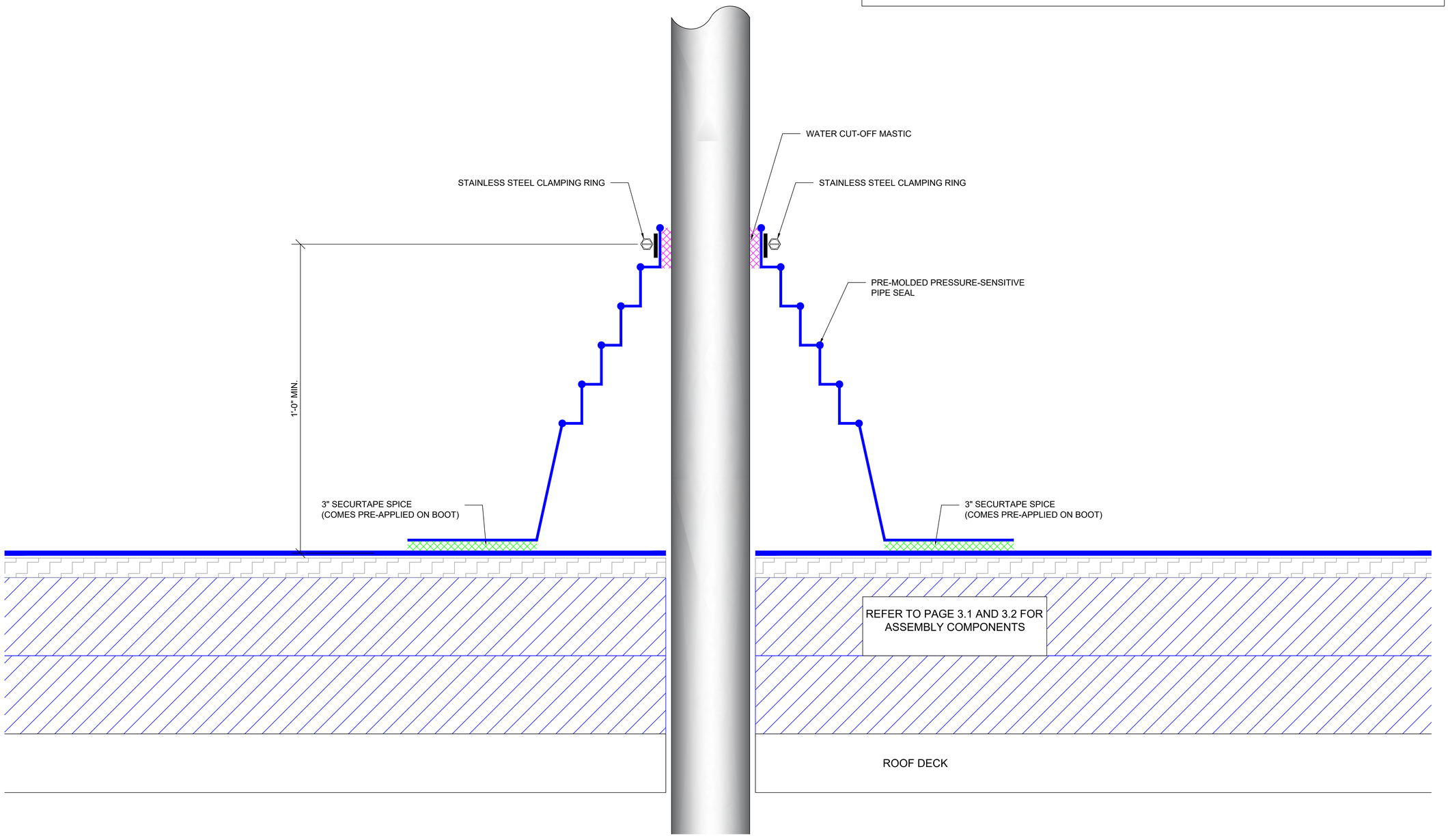
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GENERAL NOTES:

-PIPES TO BE FREE OF ALL DEBRIS PRIOR TO INSTALLATION OF PIPE FLASHING AND WATER CUT OFF MASTIC.

-PIPES TO BE SET BY PLUMBER TO GIVE AMPLE HEIGHT FOR FLASHING. PLUMBER IS RESPONSIBLE FOR SECURING PIPE STACK TO STRUCTURE.

-TAPERED INSULATION NOT SHOWN FOR CLARITY.



PIPE

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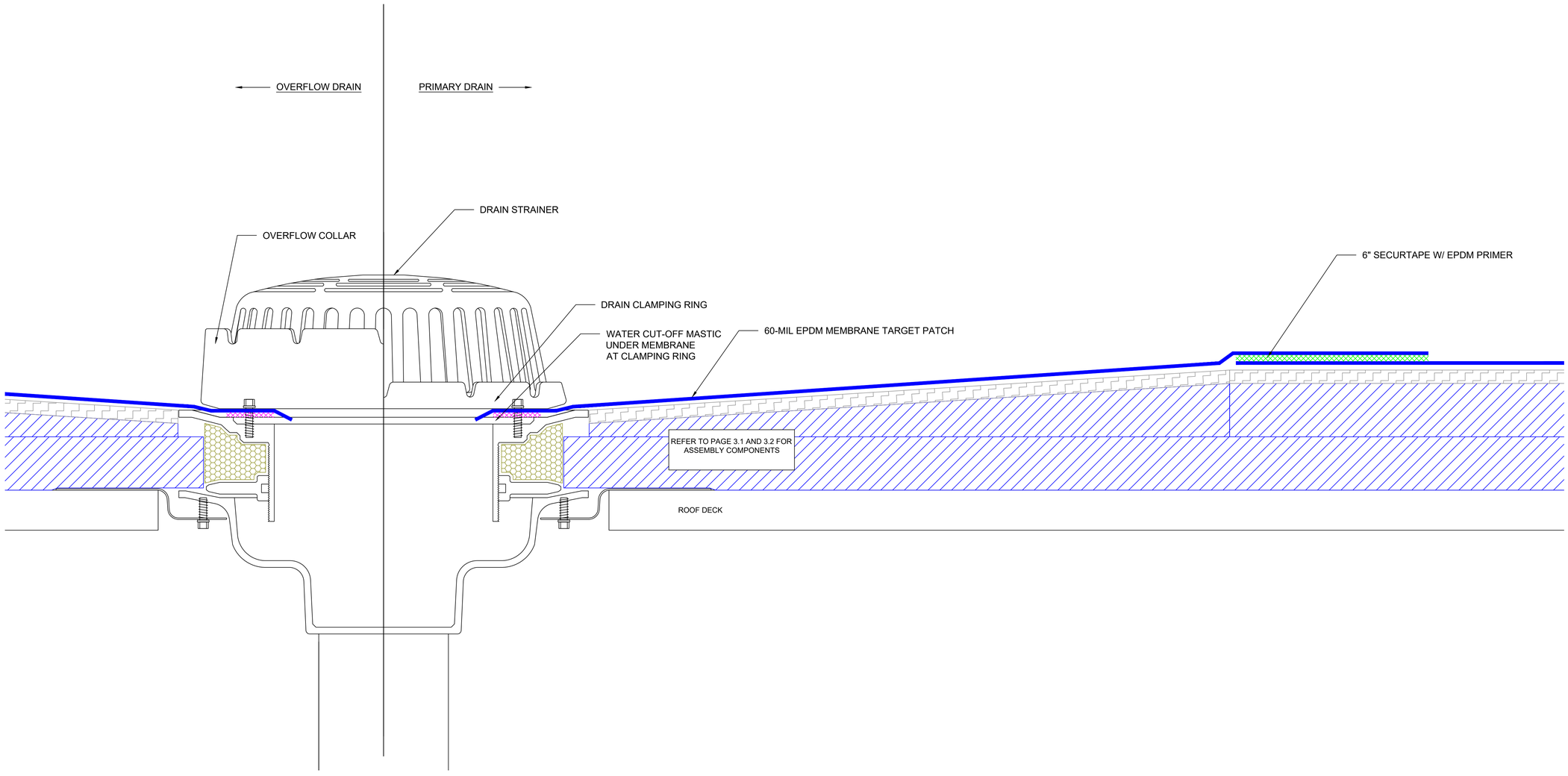
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DRAIN

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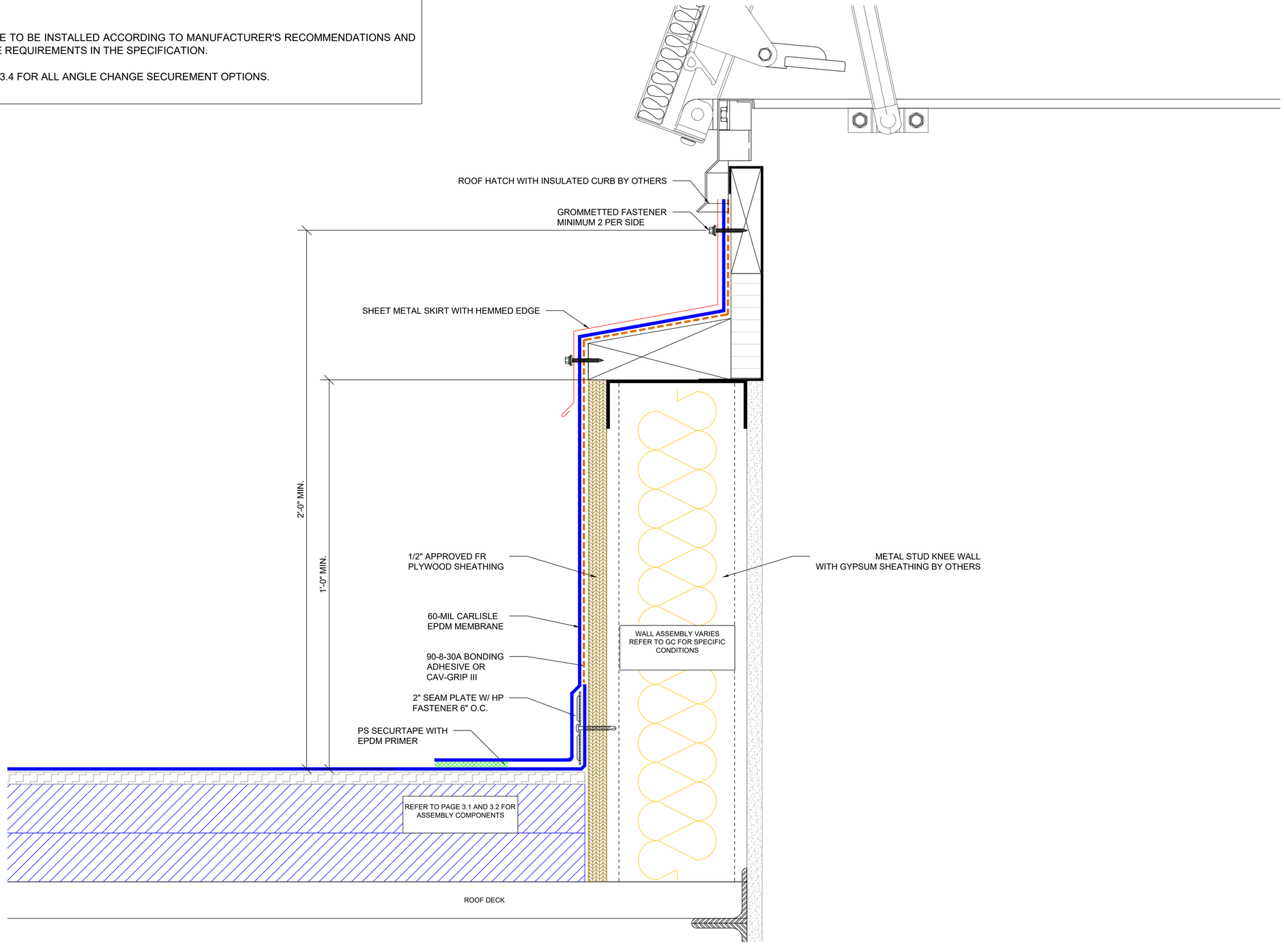
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ROOF HATCH

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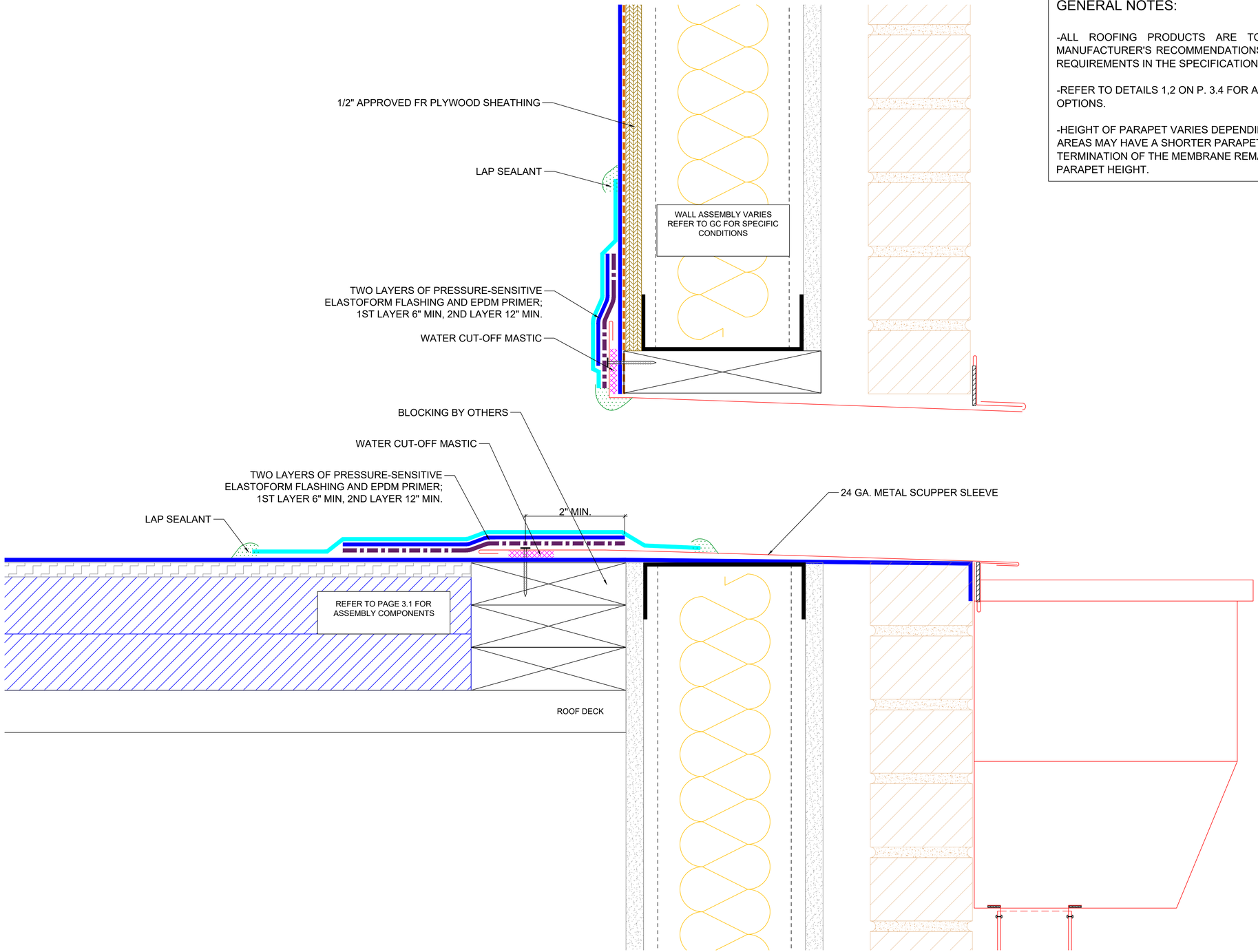
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SCUPPER

REF. ARCH. DET.: 16/A4.2

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-HEIGHT OF PARAPET VARIES DEPENDING ON LOCATION OF DETAIL. SOME AREAS MAY HAVE A SHORTER PARAPET HEIGHT THAN WHAT IT DEPICTED. TERMINATION OF THE MEMBRANE REMAINS THE SAME INDEPENDENT OF PARAPET HEIGHT.

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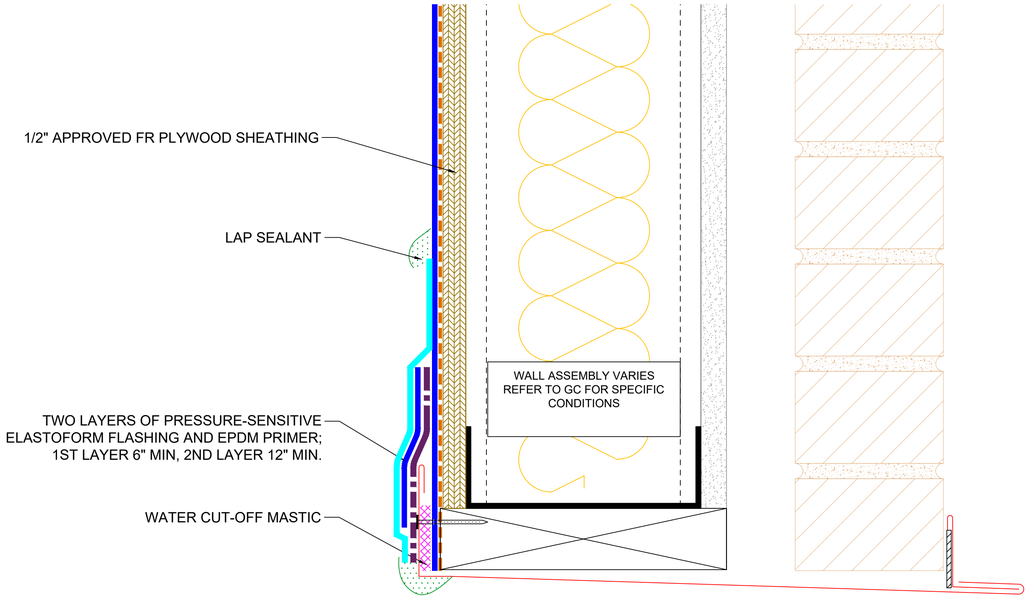
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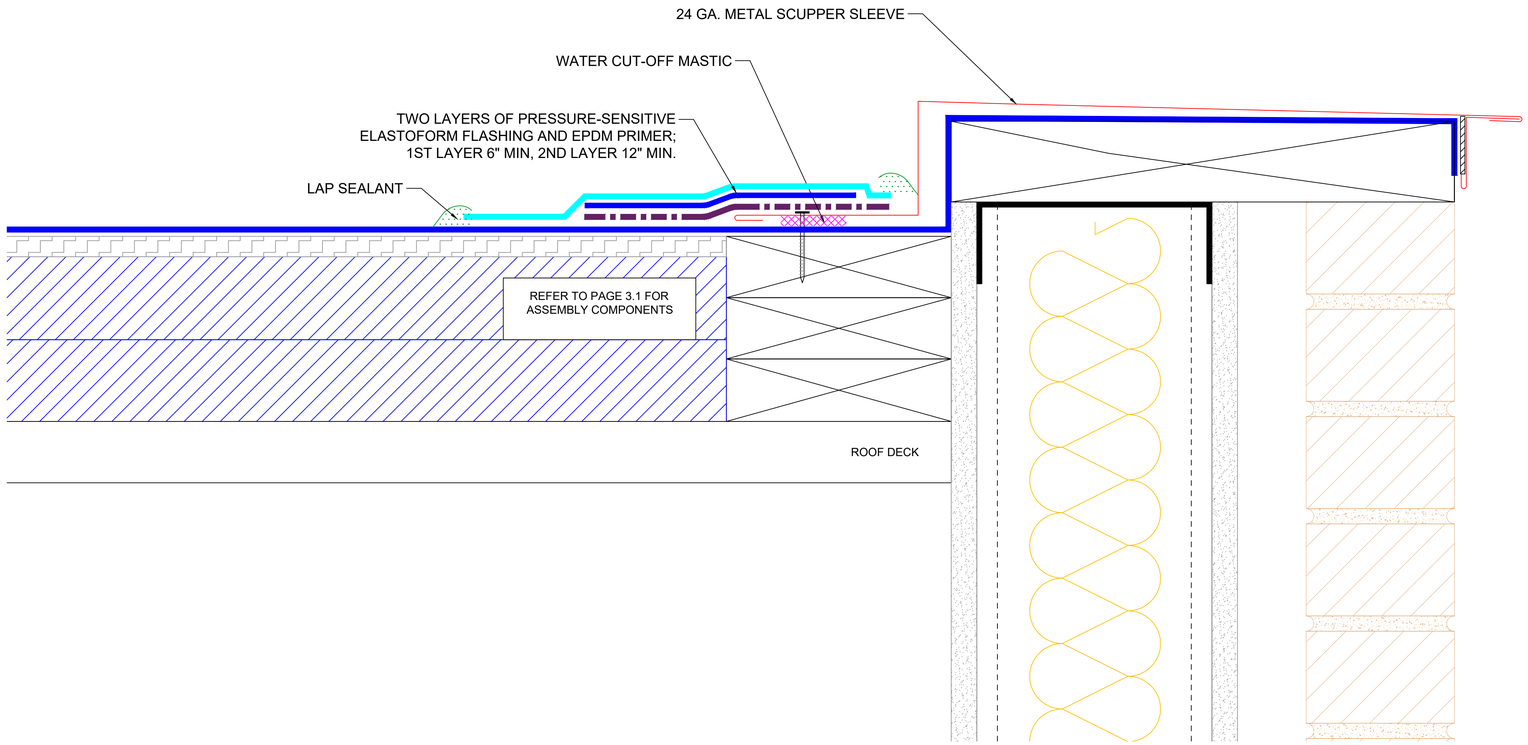
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OVERFLOW SCUPPER

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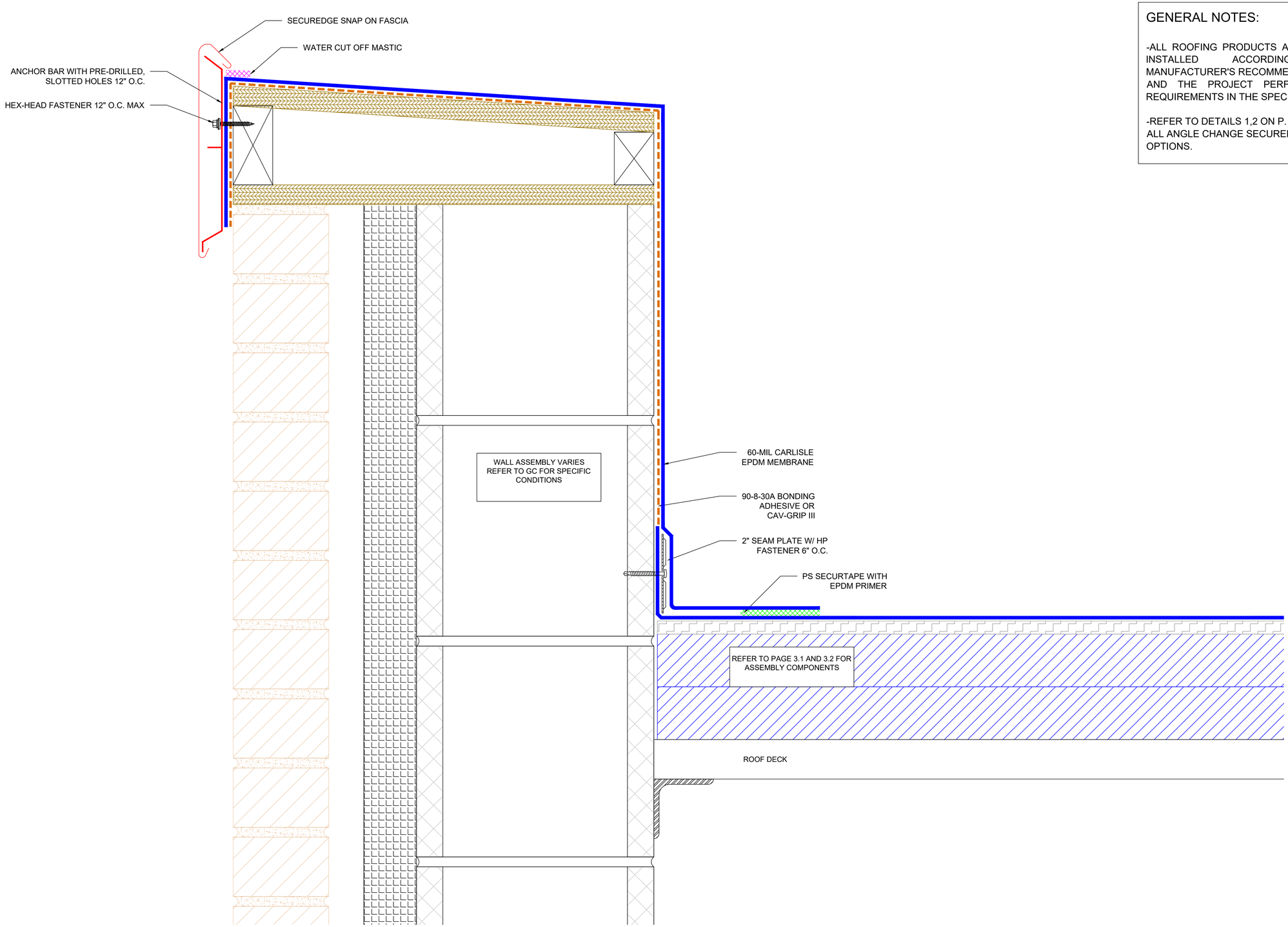
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BRICK PARAPET

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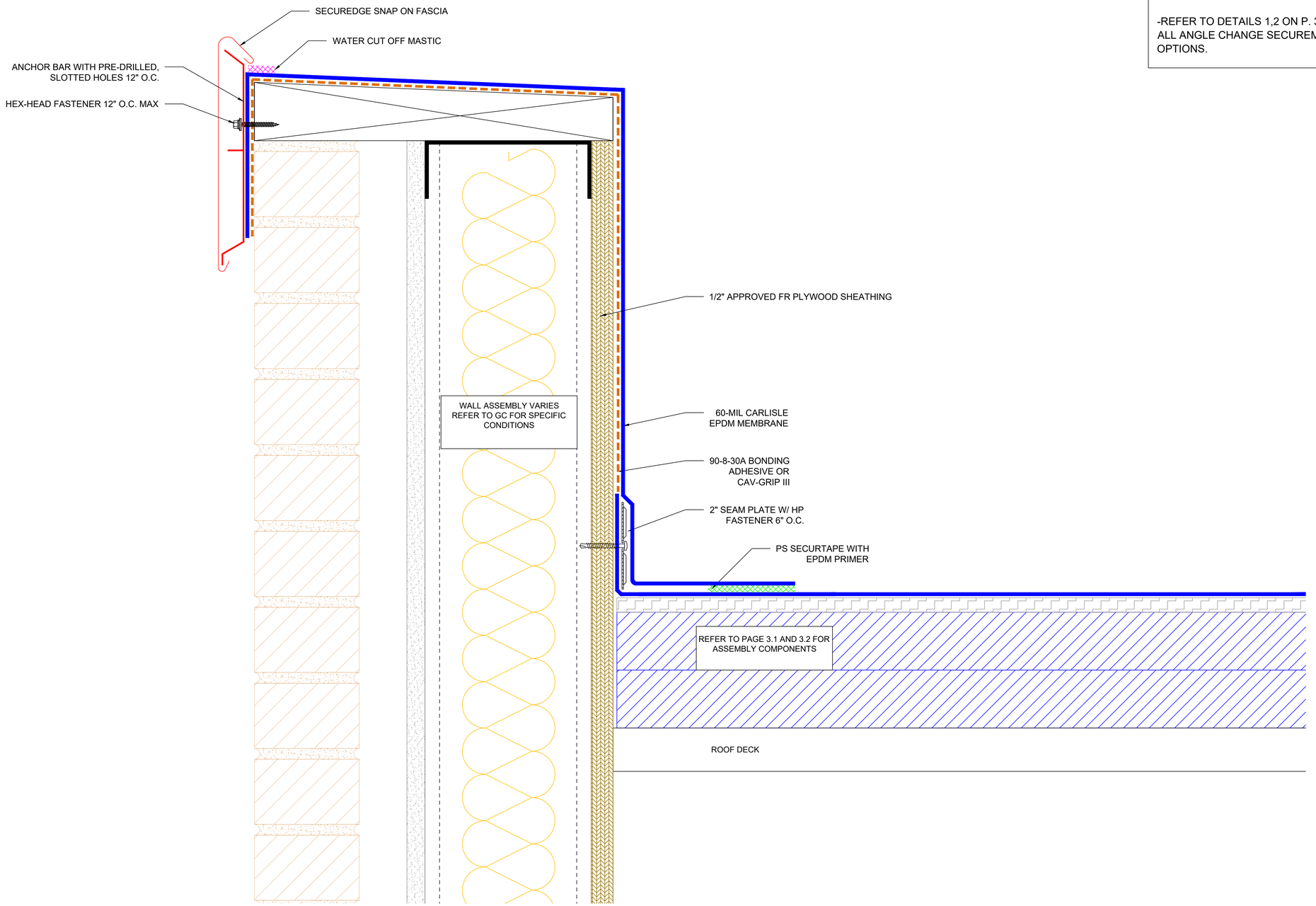
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STUD PARAPET

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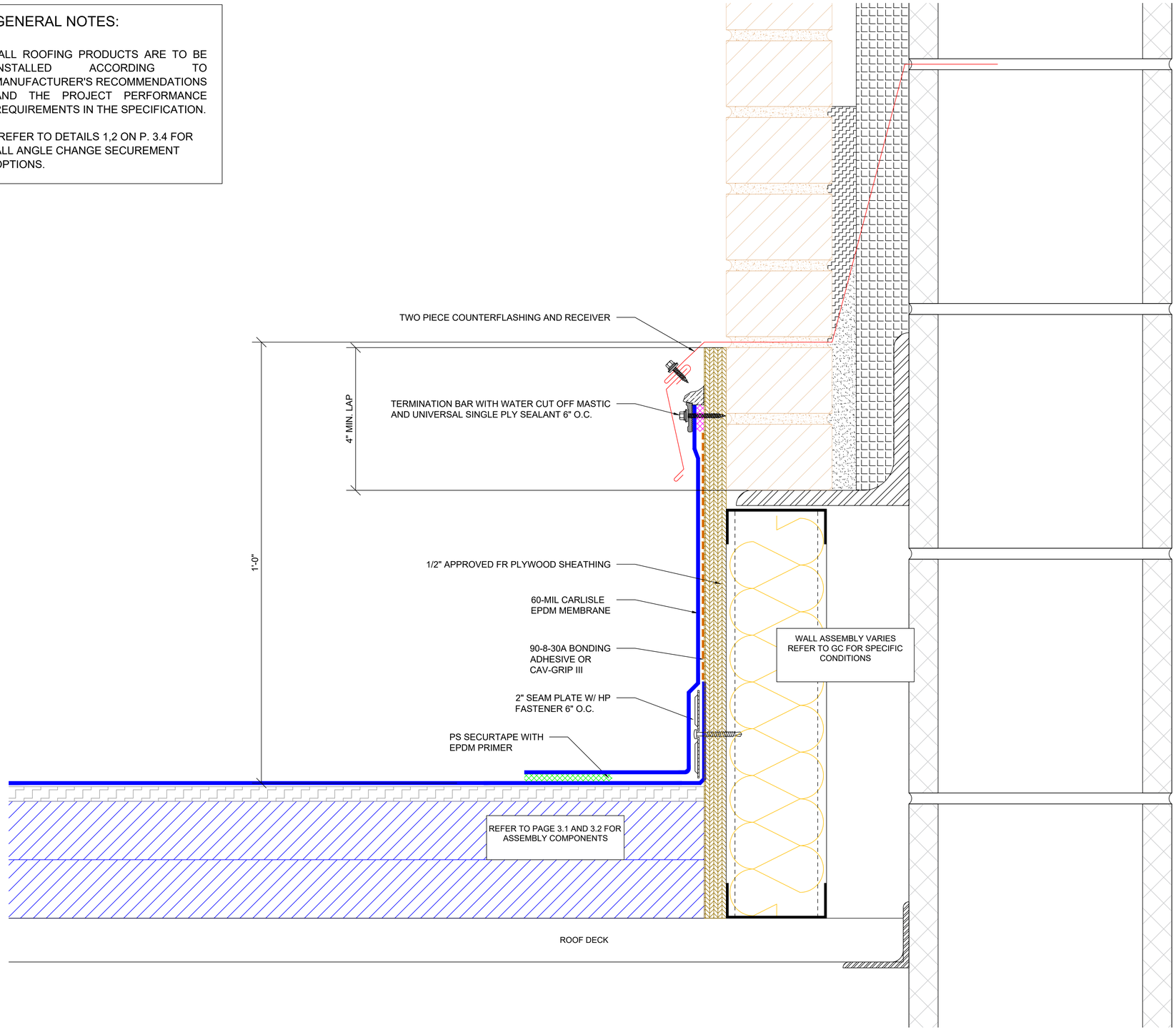
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RISEWALL

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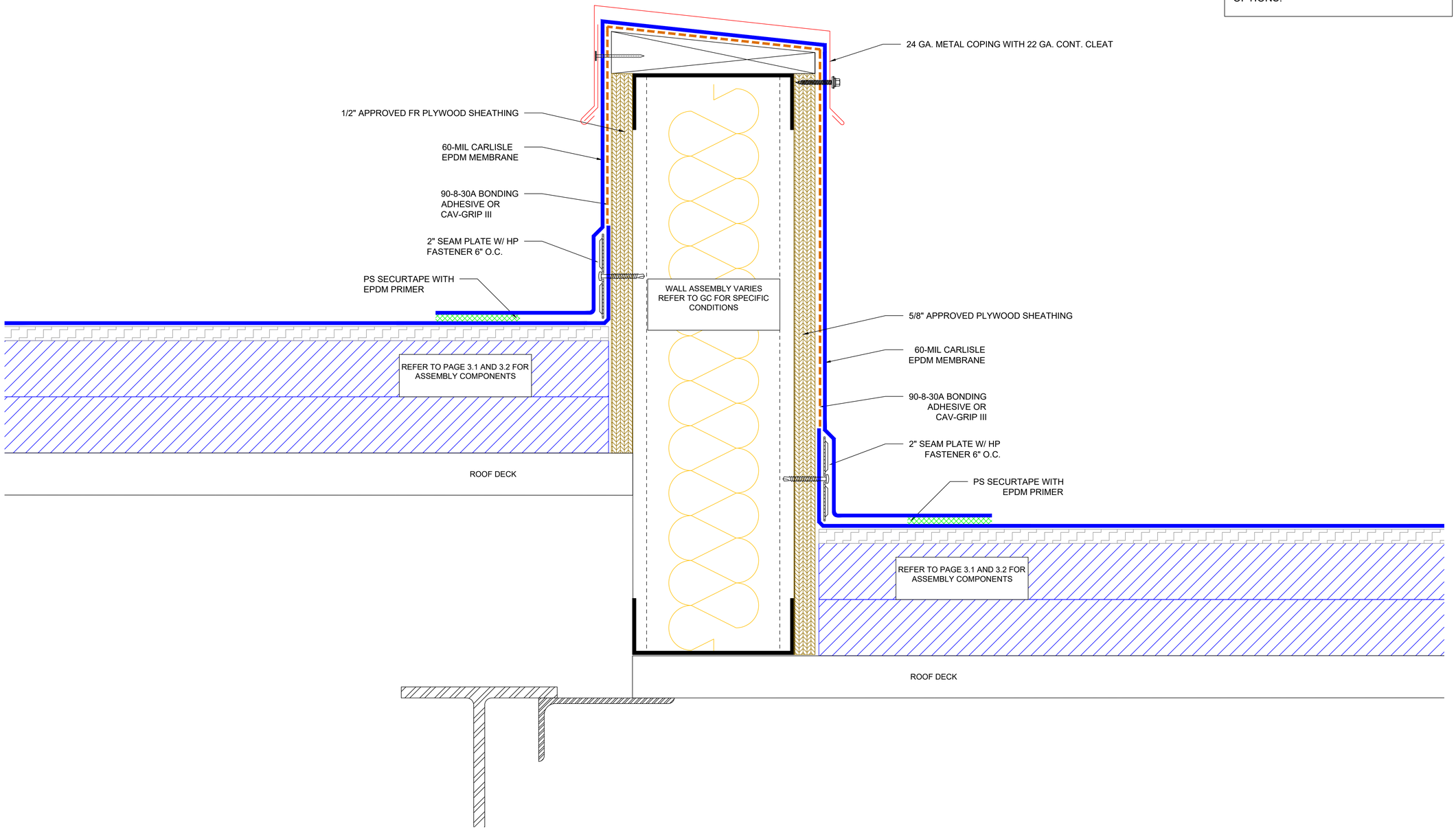
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AREA DIVIDER

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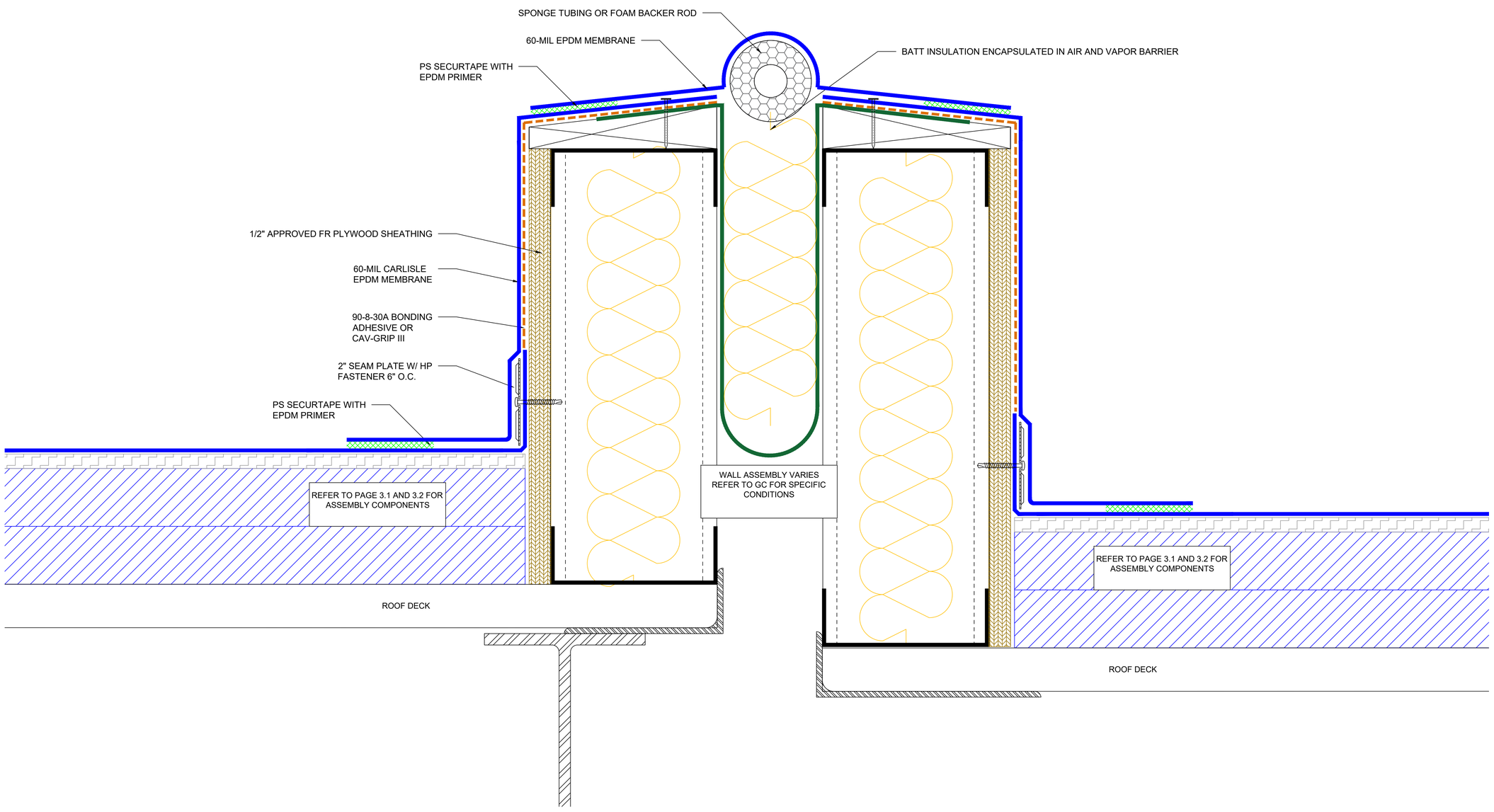
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ROOF TO ROOF EXPANSION JOINT

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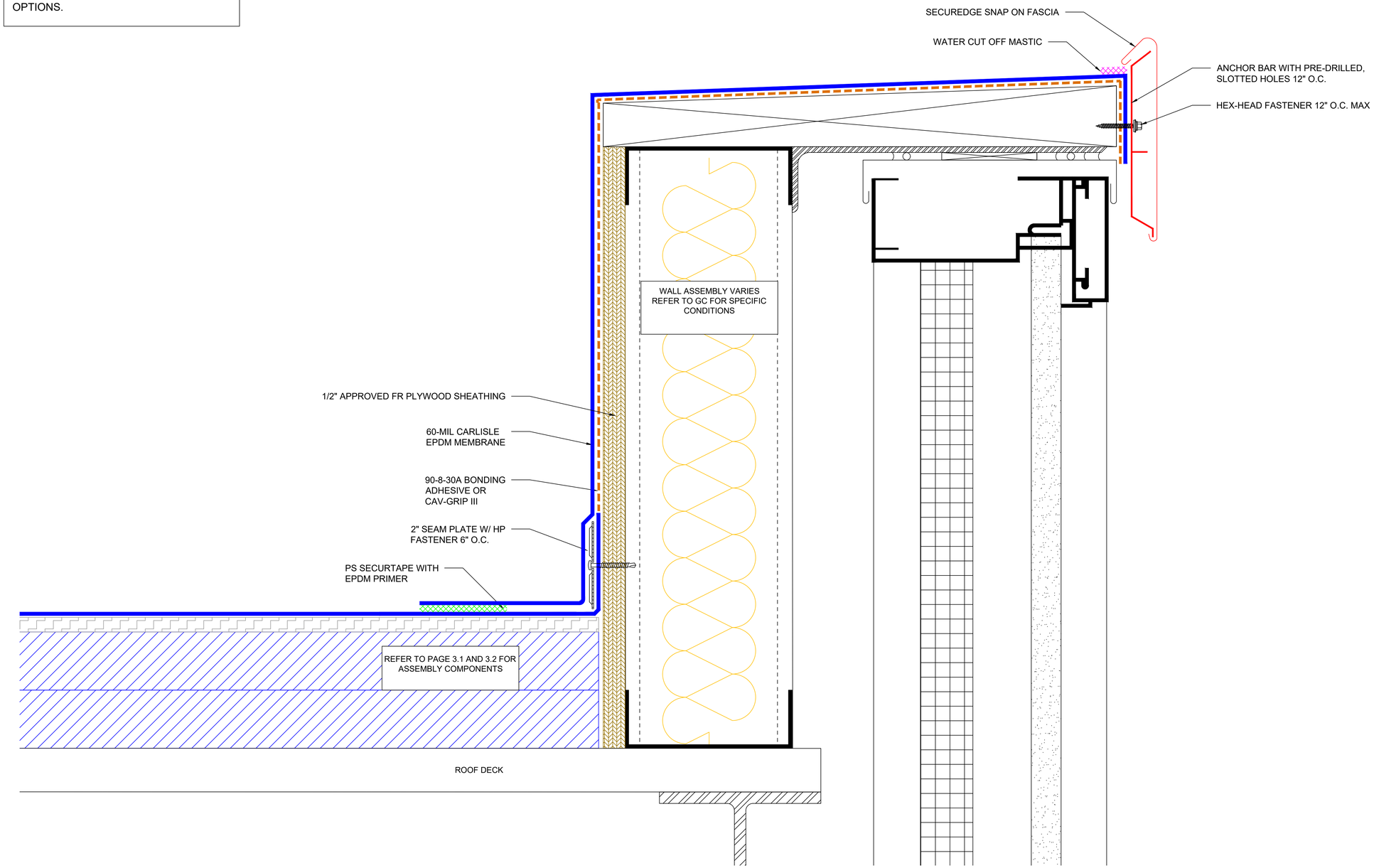
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ALUM. FRAME HEAD

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